

AWS Roadmap: Beginner → Advanced (Complete Guide)

(Perfect for DevOps, Cloud, and beginner engineers planning a strong foundation in AWS)

3-Month AWS Learning Timeline (Beginner → Advanced)

Month 1 — AWS Foundations (Core Concepts + Essential Services)

Goals: Build fundamentals, understand core AWS services, learn console basics, do small hands-on tasks.

Week 1: Cloud Basics + IAM

- What is Cloud Computing? IaaS, PaaS, SaaS
- AWS Global Infrastructure (Regions, AZs, Edge Locations)
- IAM Basics: Users, Groups, Roles, Policies
- MFA setup, AWS CLI installation

Hands-on:

- Create free-tier account
- Create IAM User + Role
- Attach policies (S3FullAccess, EC2ReadOnly, custom least-privilege policy)

Week 2: Compute Essentials

- EC2: AMI, Instance Types, Key Pairs, Security Groups
- Auto Scaling Groups (ASG)
- Elastic Load Balancer (ALB, NLB basics)

Hands-on:

- Launch EC2 Linux machine
- SSH into instance
- Install Apache/Nginx
- Create ALB + ASG mini setup

Week 3: Storage Fundamentals

- S3: Buckets, Versioning, Lifecycle Rules, Glacier
- EBS vs EFS
- S3 static website hosting

Hands-on:

- Host static website on S3
- Create lifecycle rule to move objects to Glacier
- Attach EBS volume to EC2

Week 4: Networking Basics

- VPC: CIDR, Subnets, Route Tables
- NAT Gateway, Internet Gateway
- Security Groups vs NACLs

Hands-on:

- Create a custom VPC
 - Create public + private subnets
 - Deploy EC2 in both and test internet access
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Month 2 — Intermediate AWS (Networking, Databases, Security, Monitoring)

Week 5: Databases

- RDS (MySQL/Postgres)
- DynamoDB basics
- Aurora overview

Hands-on:

- Deploy RDS MySQL
- Connect from EC2 using MySQL CLI
- Create DynamoDB table

Week 6: Security + Management

- KMS basics
- CloudTrail vs CloudWatch
- CloudWatch Logs, Metrics, Alarms
- AWS Config overview

Hands-on:

- Create custom CloudWatch dashboard
- Create CloudWatch alarm for EC2 CPU
- Enable CloudTrail

Week 7: Serverless

- Lambda basics
- API Gateway
- EventBridge
- SNS + SQS

Hands-on:

- Create Lambda function (Python/Node.js)
 - Trigger via API Gateway
 - Build SQS + Lambda workflow
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Week 8: Containers on AWS

- ECS Basics
- ECR
- Fargate
- Introduction to EKS (Kubernetes)

Hands-on:

- Push Docker image to ECR
 - Deploy service on ECS Fargate
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Month 3 — Advanced AWS (Architecting + DevOps + Automation)

Week 9: AWS DevOps Tools

- CodeCommit
- CodeBuild
- CodePipeline
- CI/CD on AWS
- CloudFormation basics

Hands-on:

- Build CI/CD pipeline for an S3 static website OR Lambda

Week 10: High Availability & Architecture

- Multi-AZ vs Multi-Region
- Disaster Recovery patterns
- Elasticity & Scalability
- Well-Architected Framework (5 Pillars)

Hands-on:

- Design and diagram a 3-tier architecture
- Multi-AZ RDS + ASG setup

Week 11: Automation + Advanced CloudFormation

- YAML basics
- Infrastructure as Code best practices
- CDK overview

Hands-on:

- Deploy EC2 + S3 via CloudFormation template

Week 12: Final Prep + Advanced Topics

- AWS cost optimization
- Spot Instances

- Data Engineering on AWS (Glue, Athena, Lake Formation)
- Security best practices

Hands-on:

- Create cost alerts
 - Run Athena queries on S3 data
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Best Free Resources to Learn AWS

Documentation & Guides

- AWS Skill Builder (Free Tier Courses)
- AWS Workshops (best hands-on experience)
- AWS Well Architected Labs

Practice Labs (Free/Paid)

- AWS Cloud Quest (Free for beginners sometimes)
- Qwiklabs (now Google Cloud Skills Boost—still has multi-cloud labs)
- KodeKloud AWS Labs

Free Practice Exams

- Tutorial Dojo (free section)
- Whizlabs free tests
- ExamPro YouTube practice exams

Top 5 YouTube Channels to Learn AWS

1. [Stephane Maarek](#) (AWS Guru)
 2. [freeCodeCamp AWS courses](#)
 3. [Abhishek Veeramalla](#) (AWS zero to hero)
 4. [Trainwithshubham](#)
 5. [Gaurav Sharma](#)
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Best AWS Hands-On Projects (Beginner → Advanced)

Beginner Projects

1. Host a Static Website on S3
2. Deploy EC2 + Apache Web Server
3. Build Multi-tier VPC with public/private subnets

Intermediate Projects

1. **CI/CD Pipeline using CodePipeline + CodeBuild + S3**
2. **Serverless URL Shortener (Lambda + API Gateway + DynamoDB)**
3. **ECS Fargate Microservice Deployment**

Advanced Projects

1. **Hybrid Cloud Architecture with VPN between On-prem & AWS**
 2. **Multi-Region Disaster Recovery Setup**
 3. **Data Lake on AWS (S3 + Glue + Athena)**
 4. **CloudFormation IaC Project for complete VPC + EC2 + RDS stack**
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How to Practice AWS Step-by-Step (Safely & Effectively)

Step 1: Create Free Tier Account

- Use a fresh email
 - Add credit/debit card
 - Enable billing alerts
 - Set AWS Budgets with thresholds (₹500, ₹1000 alerts)
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Step 2: Set Your Workspace

Install:

- AWS CLI
 - VS Code
 - Terraform (optional)
 - Git
 - Node.js/Python
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Step 3: Practice in This Order

1. Start with S3
 2. Deploy EC2
 3. Create VPC
 4. Add DB (RDS)
 5. Add Autoscaling + Load Balancers
 6. Move to Lambda & API Gateway
 7. Try containers (ECR, ECS)
 8. Build CI/CD pipeline
 9. Deploy with CloudFormation
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Step 4: Document Everything

Maintain a personal GitHub repo:

- `aws-notes`
- `aws-projects`
- `cloudformation-templates`

This becomes resume gold.

Step 5: Monitor + Clean Up Resources

After every lab:

- Delete EC2 instances
 - Delete EBS volumes
 - Delete RDS
 - Delete ALB/NLB
 - Empty and delete S3 buckets
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Final AWS Roadmap Diagram (Text Form)

Beginner:

Cloud Basics → IAM → EC2 → S3 → VPC → Basic Hands-on

Intermediate:

RDS → CloudWatch → Lambda → API Gateway → SNS/SQS → ECS → CI/CD

Advanced:

CloudFormation → High Availability → Multi-Region → Architectures → Cost Optimization → Data Engineering
